

Decision Tool

Glider Selection Dashboard

“Help Me Decide”

PROJECT TEAM & CONTRIBUTORS

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Background/Justification



Glider Selection Dashboard

Develop a Glider Selection Dashboard that brings together information from the Glider Rodeo, glider manufacturers, and relevant datasets to help users determine which glider platform and configuration would be best suited for a particular mission.



Interactive Decision-Support

Learn more about glider options available and the different PAM systems. Move beyond simply providing a catalogue of available gliders and instead create an interactive decision-support tool.



Methods

- Brainstorm dashboard features as a group
- Data obtained from manufacturer
- Selection tools created in Quarto hosted via GitHub pages
 - Started with NOAA PAM template
 - Python — pandas, openpyxl, xarray, netCDF4
 - JavaScript — Plotly (2-D and gl3d 3-D), Leaflet + Leaflet.draw (maps), Turf.js (geometry)
- Leveraged AI tools for website design
- Collaborative feedback & editing
- External data and services
 - GMRT Synthesis — bathymetry fetched live for any drawn polygon
 - Esri World Ocean Base — basemap tiles
 - Glider Playground (Orlando Prugel-Bennett, National Oceanography Centre, Apache-2.0) — the 3-D seafloor view follows its approach

Use of AI



ChatGPT was used for Study Design Tool, Debugging



BY ANTHROPIC

data extraction, analysis code, plotting, the website itself



Gemini was used to create visual for presentation slides.

Results

[Link](#)



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Glider Selection Dashboard

Comparing the PAM-equipped gliders of the [NOAA Fisheries Glider Rodeo 2026](#). Same structure Kasey built — set what the mission needs, apply, then your options — with photos, sourcing and the rodeo measurements added.

Mission Parameters

Enter requirements for your mission below. Tick a row to make it a requirement; leave it unticked and it is ignored. Then press Apply.

BATTERY TYPE

☒ Any ☐ Rechargeable ☐ Primary (single-use)

MISSION PARAMETERS

☐ Deployment duration 30 days ☐ Depth range 50-300 m ☐ Speed needed 1.0 kt
1 738 5 1,500+ 0.2 3.5
☐ Maximum weight 80 kg
20 200

☐ Deployment / recovery capability

☒ Single-person handling ☐ Two-person handling ☐ Crane / LARS available

Choose the highest level of handling support available. Higher capability also includes gliders with lower handling requirements.

SCIENCE PAYLOAD

Pick what you need to measure, then a specific sensor if you have one in mind. The list narrows to gliders that sensor is already integrated on. Sensors and their integrations come from the glider/PAM workbook.

MEASUREMENT

Any

SENSOR

Any sensor

PAM SYSTEM

Choose a recorder to keep only the gliders it is documented on, and to see its memory, channel count and bandwidth.

Study Design Tool

Draw a study area, choose a simple transect design, and estimate survey distance and nominal mission duration.
This prototype is intended for first-order study design rather than operational mission planning.

Results

[Link](#)

Study requirements

Use the polygon tool on the map to define one study area. Edit the polygon at any time; the design will update when you regenerate it.

SURVEY GEOMETRY

Survey pattern

Lawnmower (connected transects) ▼

Transect spacing 10 km

Track orientation 0°

0° = north-south transects; 90° = east-west.

MISSION ASSUMPTIONS

Nominal horizontal speed 0.50 kt

Glide angle 26°

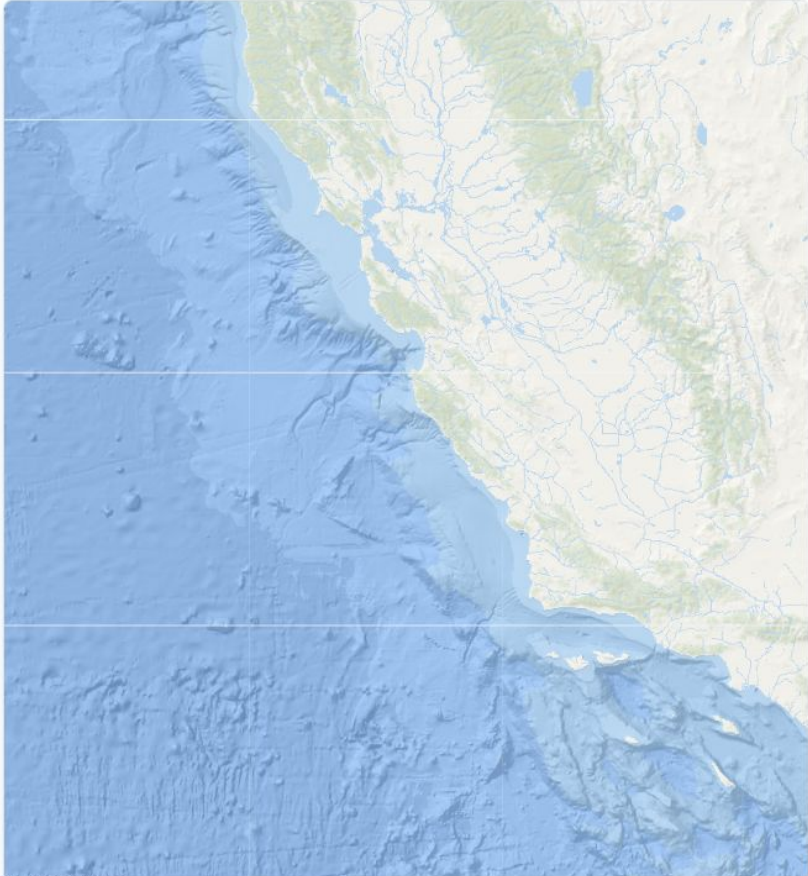
Angle from horizontal. Used with target depth to estimate the horizontal distance of one down-and-up dive cycle.

Target depth 500 **Number of gliders** 1

Draw study area Finish polygon

Clear area Generate survey

Clear tracks Load example



Loading land mask...

Results

[Link](#)

Slocum Endurance Calculation

Predicts how long a Slocum will last on a given mission, from Teledyne Webb Research's **Slocum Endurance Tool** (revision C, 2012-10-19). The workbook's formulas are evaluated here as-is rather than reimplemented, so the numbers are Excel's: all 998 cells the model computes are checked against the workbook's own stored results.

75 days
ENDURANCE (DAYS)

1,500 km
APPROXIMATE DISTANCE (KM)

Glider + standard sensors **5.77** Ah/day Total with thruster + extras **7.37** Ah/day

Mission and configuration

MISSION

Maximum profile depth m

Minimum profile depth m

Horizontal velocity m/s

Vertical velocity m/s

Surface on

Number of Profiles before surface

Surface time min

Average operating temperature °C

GLIDER CONFIGURATION

Buoyancy pump

Glider type

Altimeter ☒

Pinger ☐

FreeWave ☒

PTT / Argos ☐

BATTERY

Chemistry

Type

Extended energy bay ☐

STANDARD SCIENCE PAYLOAD

CTD

Motherboard

Persistor

Aanderaa optode

Explorer 500 kHz DVL

FLBBD-SLK

Biospherical QSP-2150

THRUSTER AND EXTRA SENSORS

Thruster* Ah/d %

UVP6 Ah/d

EK80 Ah/d

JASCO OceanObserver (1 channel)* Ah/d

JASCO OceanObserver (5 channel)* Ah/d

eDNA Ah/d

From [content/data/extra-sensors.csv](#) — add a row to add a sensor.

Results

[Link](#)

Glider Performance

Every glider in the rodeo flew the same plan: a shakedown, then **fast**, **slow**, **intermediate** and **drift** blocks, then a short repeat of slow, intermediate and fast at the end (the mini-modes, right of the hairline in each panel). Because the modes are shared, the differences below are differences between the *platforms*, not between missions. Numbers come from `output/glider_performance_summary.xlsx`; each panel keeps its own y-axis, because a watt and a cubic centimetre do not belong on one scale.

GLIDERS ☒ sg597 ☒ sg274 ☒ rizzo ☒ stenella ☒ capex987 ☒ SEA117 ☒ belladonna ☐ All ☐ Show flagged values

MODES ☒ shakedown ☐ fast ☐ slow ☐ intermediate ☐ drift ☐ mini-slow ☐ mini-intermediate ☐ mini-fast ☐ All ☐ Main plan ☐ Mini repeat

METRICS ☒ Mean power draw W ☒ Energy per dive Wh ☐ Energy per km travelled Wh/km ☒ Vertical speed m/s ☒ Speed over ground km/h

☐ Distance per dive km ☐ Dive duration h ☐ Dive depth m ☒ Buoyancy pump volume cc/h ☐ Pump volume per dive cc ☐ Pump adjustments per hour

☐ Dives completed in mode count

Mean power draw W



No data: SEA117.

Energy per dive Wh



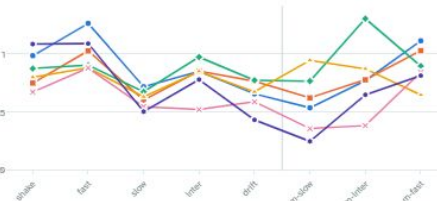
No data: SEA117.

Vertical speed m/s



belladonna is not plotted: CTD timestamps arrive in bursts (samples under 1 ms apart), so rates computed between samples are far too high. Shown for completeness only.

Speed over ground km/h



No data: SEA117.

Lessons Learned

Next Steps

Endurance calculation for Seaglider

Study design tool needs attention, testing, and feedback

Import and Export waypoints from the Study Design Tool for glider missions

Import data to the data viewer from additional glider missions.

Acknowledgements

Shannon Rankin

Openscapes

NOAA Fisheries

Selene Fregosi

Glider Rodeo Team

Aquaview

Kourtney Burger

Glider Rodeo Funders



Anne Simonis



Kasey Castello



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Schedule - time limits for each slide

Intro: 1 min

Background/Justification: 2 min

Methods: 2 min

Use of AI: 1 min

Results: 5 min

Lessons Learned: 2

Next steps: seaglider endurance calculator

Acknowledgements: 1